[leetcode.com /problems/copy-list-with-random-pointer/description/](https://leetcode.com/problems/copy-list-with-random-pointer/description/)

Unknown Title

Description

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Code

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Testcase

Testcase

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Test Result

Test Result

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[138. Copy List with Random Pointer]

Medium

Topics

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Hint

A linked list of length `n` is given such that each node contains an additional random pointer, which could point to any node in the list, or `null`.

Construct a **deep copy** of the list. The deep copy should consist of exactly `n` **brand new** nodes, where each new node has its value set to the value of its corresponding original node. Both the `next` and `random` pointer of the new nodes should point to new nodes in the copied list such that the pointers in the original list and copied list represent the same list state. **None of the pointers in the new list should point to nodes in the original list**.

For example, if there are two nodes `X` and `Y` in the original list, where `X.random --> Y`, then for the corresponding two nodes `x` and `y` in the copied list, `x.random --> y`.

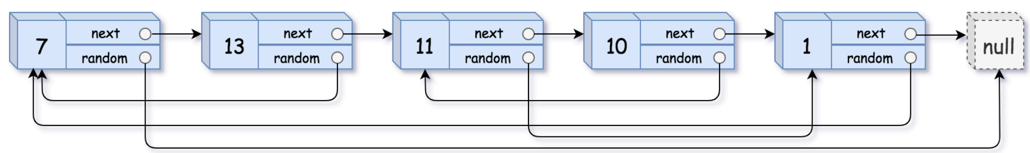
Return *the head of the copied linked list*.

The linked list is represented in the input/output as a list of `n` nodes. Each node is represented as a pair of `[val, random_index]` where:

- `val`: an integer representing `Node.val`
- `random_index`: the index of the node (range from `0` to `n-1`) that the `random` pointer points to, or `null` if it does not point to any node.

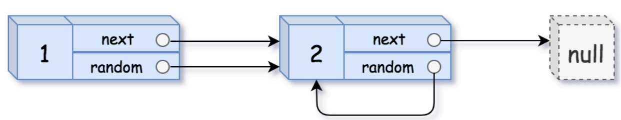
Your code will **only** be given the `head` of the original linked list.

Example 1:



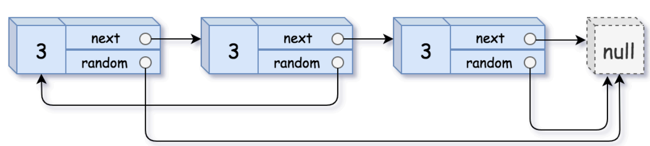
Input: head = [[7,null],[13,0],[11,4],[10,2],[1,0]]
Output: [[7,null],[13,0],[11,4],[10,2],[1,0]]

Example 2:



Input: head = [[1,1],[2,1]]
Output: [[1,1],[2,1]]

Example 3:



Input: head = [[3,null],[3,0],[3,null]]
Output: [[3,null],[3,0],[3,null]]

Constraints:

- `0 <= n <= 1000`
- $-10^4 <= Node.val <= 10^4$
- `Node.random` is `null` or is pointing to some node in the linked list.

Seen this question in a real interview before?

1/6

Yes

No

Accepted

1,937,457/3.1M

Acceptance Rate

63.1%

Companies

Hint 1

Just iterate the linked list and create copies of the nodes on the go. Since a node can be referenced from multiple nodes due to the random pointers, ensure you are not making multiple copies of the same node.

Hint 2

You may want to use extra space to keep old_node ---> new_node mapping to prevent creating multiple copies of the same node.

Hint 3

We can avoid using extra space for old_node ---> new_node mapping by tweaking the original linked list. Simply interweave the nodes of the old and copied list. For example: Old List: A --> B --> C --> D InterWeaved List: A --> A' --> B --> B' --> C --> C' --> D --> D'

Hint 4

The interweaving is done using next pointers and we can make use of interweaved structure to get the correct reference nodes for random pointers.

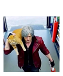
Discussion (336)

💡 Discussion Rules

1. Please don't post **any solutions** in this discussion.

2. The problem discussion is for asking questions about the problem or for sharing tips - anything except for solutions.

3. If you'd like to share your solution for feedback and ideas, please head to the solutions tab and post it there.



Antinormanist

Mar 25, 2024

i'm gonna work in McDonald's

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1.3K

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[Vaibhav Raj Goel]

Jul 25, 2020

This same question appeard in my amazon round 1.
Let me write some more about what was attending its round 1 felt.
Amazon round 1 contained 4 section.
1. 7 Array based Question of debugging came. (Which were so easy that everyone solved them with all test cases as passed). (20 min)
2. This section contained 2 Coding based Questin were given, 1 was 138. Copy List with Random Pointer and other question was to return a recommendataion gener list of book based on some information given, also very simple. But the time was of constraint here. (60 min)
3. This section had some Behavioural question.(answers vary person to person).
4. last section was combination of Technical and Apti Questions (25 Question in 34 min).

That's it I hope it come of some use to you.

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[Arun Giri]

Apr 22, 2020

What does the question asks?
I was confused with the question for a bit. But understood it, Basically the question is to deep copy the linked list, that is make an exact copy.(replicate the random nodes too)
If it was simple single linked list, it was simple. Just make a copy on the go and connect the nodes.

But if the random element is present we have to replicate it to, just imagine you made a new node and the random of this node points to a node which is not yet created ?What will you do? How will you Point to that node, thats what we have to solve here,

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tonyraubenheimer

Jun 08, 2020

The question description indicates that node.random is an index. But, the inputs are structured such that node.random is actually a pointer.

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[napoleon]
Nov 23, 2013

I solve this problem by costing 392ms.
I use map to save the relation between the original list and the copy one.

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SIRISH CHANDRAA DHANANJAYA
Jul 22, 2018

The problem description is not clear.

(1) What is meant by deep copy of a linked list?
(2) Why is there no sample input/output?
(3) Why is there no "Run Code" option with sample test case?

Please fix.

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ufo2mstar

Oct 04, 2018

It seems like the problem is merely asking for a deepcopy of the labels (node values) of the random node that nodes point to, not the absolute reference of the n-th node within the Linked List!

I was mapping the internal relative index of the random destination node and was preserving the order,
but the silly validation program is passing even if I create brand new references!

I feel that either the problem should be restated to make this clear,
or the test cases should look into the node addresses properly!

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Jason Lu
Nov 09, 2013

Does anyone know what

Input: {1,2,2,2}
Output: {1,2,#,#}
Expected: {1,2,2,2}

means?

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Reply

raz

Aug 29, 2018

I dont understand the qn. can someone plz explain.
thanks

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pengtsen

Nov 12, 2019

I have coded the O(n) space solution before and never thought I can do better. I recently saw this question in an interview and gave the o(n) space solution, but it is deemed unaccetable. You are supposed to give a O(n) time and O(1) space solution.

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83 Online

```
}
*/
class Solution {
    public Node copyRandomList(Node head) {
        return head;
    }
}
```

Saved

Ln 18, Col 21

Wrong Answer

Runtime: 0 ms

Diff

Case 1

Case 2

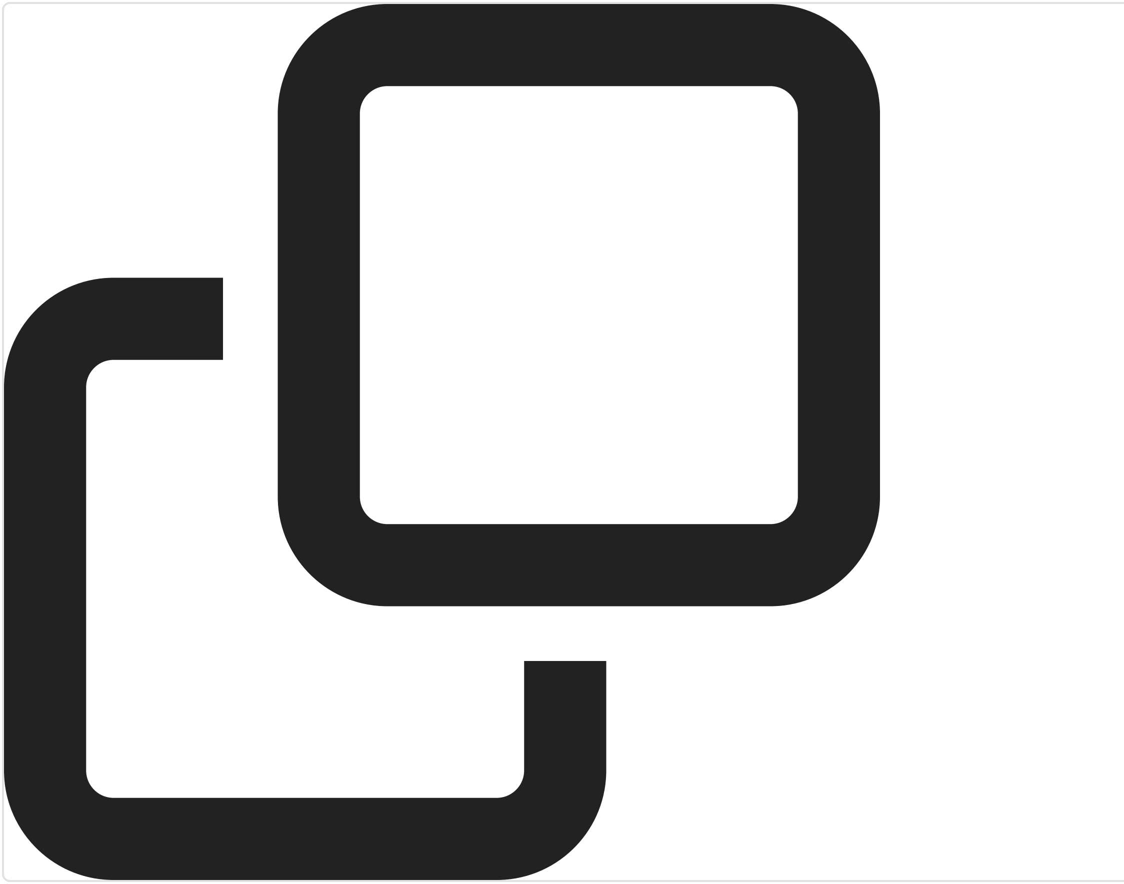
Case 3

Input

head =

[[7,null],[13,0],[11,4],[10,2],[1,0]]

Output



Node with label 7 was not copied but a reference to the original one.

Expected

[[7,null],[13,0],[11,4],[10,2],[1,0]]

Contribute a testcase

Input

9

1

2

3

›

[[7,null],[13,0],[11,4],[10,2],[1,0]]

[[1,1],[2,1]]

[[3,null],[3,0],[3,null]]

Output

9

1

2

3

›

Node with label 7 was not copied but a reference to the original one.

Node with label 1 was not copied but a reference to the original one.

Node with label 3 was not copied but a reference to the original one.

Expected

9

1

2

3

›

[[7,null],[13,0],[11,4],[10,2],[1,0]]

[[1,1],[2,1]]

[[3,null],[3,0],[3,null]]

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